ZTD2PWV

A Manual

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1 Introduction

The idea of utilizing GNSS network for meteorological observations was first put forth by Bevis et al. [1992, 1994]. Due to atmospheric refraction, GNSS signals passing through the atmosphere experiences delays and bending. Consequently, the accuracy of observations are degraded and must be corrected. Since it is difficult to accurately model this correction, standard procedure treats it as a nuisance parameter, known as zenith total delays (ZTD). As a result, ZTD is a common product of GNSS data analysis.

Geodesists typically have no further needs for ZTD. However, ZTD contains useful information on atmospheric water vapor, one of the most important greenhouse gases. Various studies have utilized this information to study various meteorological phenomena and climate. Usually, for this purpose, ZTD is converted into precipitable water vapor (PWV).

To obtain PWV, we use the following relation:

$$PWV = \Pi \cdot ZWD \tag{1}$$

where ZWD is the zenith wet delay that can be obtained by subtracting the ZTD with ZHD:

$$ZWD = ZTD - ZHD \tag{2}$$

ZHD can accurately be determined using the formula by Saastamoinen [1972]:

$$ZHD = \frac{0.0022768 \cdot p_s}{1 - 0.00266 \cdot \cos(2\varphi) - 0.28 \cdot 10^{-6} \cdot h_{ell}}. \tag{3}$$

where p_s is in situ surface pressure values in hPa, φ is the latitude of the station, and h_{ell} is the height of the station from the ellipsoid.

The dimensionless parameter Π in Eq.(1) is given by [Askne and Nordius, 1987]:

$$\Pi = \frac{10^6}{\rho R_v \left[\frac{k_3}{T_m} + k_2'\right]} \tag{4}$$

where ρ is the density of liquid water, R_v is the specific gas constant for water vapor, k_2' and k_3 are constants, and T_m is the mean temperature weighted with water vapor pressure. Nilsson et al. [2013] state that the values of Π is around 0.16. However, it can vary by as much as 20%, depending on the location, altitude, season, and weather [Bevis et al., 1994].

Davis et al. [1985] defines T_m as:

$$T_m = \frac{\int_{h_s}^{\infty} \left(\frac{e}{T}\right) dz}{\int_{h_s}^{\infty} \left(\frac{e}{T^2}\right) dz} \tag{5}$$

where e is the partial pressure of water vapor in hPa and T is temperature in Kelvin. This formula requires the knowledge of vertical profiles of temperature and humidity, which can be obtained from either radiosonde or Numerical Weather Models (NWM). One alternative is to use a formula by Bevis et al. [1992]:

$$T_m \approx 70.02 + (0.72 \times T_s) \tag{6}$$

By using this formula, one can use surface temperature T_s to obtain T_m, since both appear to be linearly correlated. Several regional T_m-T_s models are also available, which were derived using the same principle as outlined by Bevis et al. [1992].

2 About ZTD2PWV

ZTD2PWV is free software developed in Fortran for converting ZTD to PWV. Users only need to prepare the input data in the appropriate format and run the executable file `ZWD2PWV`. The zip file contains `ZTD2PWV`, as well as `MET`, `RES`, and `TRO` directories. They all must be located in the same working directory, otherwise the program will not run. Example files in `TRO` (ZTD files) and `MET` (for meteorological files) are also included for the purpose of testing.

2.1 Installation

Since the program is given as an executable file, no installation is required. However, this program requires Linux environment to run properly. For users who use Windows, it is recommended to install Windows Subsystem Linux (WSL). At the moment, we have not tested this in MacOS. Proceed with caution. Since no Fotran libaries are used in the program, it is not necessary to have a Fotran compiler installed in your machine.

1. Obtain the `ZTD2PWV` zip file from the official distribution source.
2. Place the executable file in your project's working directory.
3. Open your WSL or Linux terminal, and make the executable file runnable with the command: `chmod +x ZTD2PWV`.
4. To check whether the program works properly, run `./ZTD2PWV`. This will process all the example files in the `TRO` and `MET`. Once it's finished, all the output files (`.pwv`) can be found in the `RES`.

2.2 Directories

These directories should already exist when you download ZTD2PWV, including some example files (ZTD and meteorological files). If not, you can create them manually.

- `TRO` contains all ZTD files in the IGS format `.TRO` or `.YYzpd` (see Section 2.3.1).
- `MET` contains all meteorological parameters in a specific format `.met` (see Section 2.3.3).
- `RES` contains all the output files in a specific format `.pwv` (see Section 2.4).

2.3 Input Files

2.3.1 ZTD

This program reads ZTD files using the standard IGS format. Please refer to `https://igs.org/products/#troposphere` to see the current directory of the files.

Before GPS Week 2237, the file names follow the format:

`SSSSDDD#.YYzpd`

with the following description:

SSSS	IGS monument name
DDD	3-digit day of year
#	file number for the day, typically 0
YY	2-digit year

Since GPS Week 2238, the file names follow the format:

`IGS0OPSFIN_YYYYDOYHHMM_01D_05M_SITENAME_TRO.TRO`

with the following description:

YYYY	4-digit year
DOY	3-digit day of year
HH	2-digit hour
MM	file number for the day, typically 0
SITENAME	9 character site name

All the files will have a gzip compressed format and must be unzipped then copied to `TRO`. Make sure that `TRO` only contains the files you want to process, since ZTD2PWV will batch run all the files in the directory.

2.3.2 Station Information

The coordinates of the stations must be compiled in the file `STA_INFO` in the `TRO` directory. This must contain station names, latitude, longitude, and ellipsoidal height in meters. Below is the content of `STA_INFO` file. From left to right: station name, latitude, longitude, and ellipsoidal height in meters. An example file is included in the `TRO` directory. Please refer to it when creating your own `STA_INFO` file.

```
AIRA   31.824061   130.599592   314.6000
BAKO   -6.491055   106.848912   158.1164
```

2.3.3 Meteorological Files

Converting ZTD to PWV requires surface meteorological parameters, primarily pressure and temperature. Typically, permanent GNSS stations are equipped with in situ meteorological sensors. If in situ meteorological data are not available, users may also get these values from other sources, e.g. numerical weather models or empirical models, such as GPT3.

Below is the content of the `.met` file. From left to right: epoch, surface pressure (in hPa), surface temperature (in degree Celcius), and weighted mean temperature (in Kelvin). Some example files are included in the `MET` directory. Please refer to it when creating your own `.met` files.

```
% yyyy:doy:sod   p(hPa)     T(C)     Tm(K)
20:097:00000   981.3140   13.6909   276.0408
20:097:00300   981.3138   13.6913   276.0412
20:097:00600   981.3135   13.6917   276.0416
20:097:00900   981.3132   13.6922   276.0420
20:097:01200   981.3129   13.6926   276.0423
20:097:01500   981.3126   13.6930   276.0427
20:097:01800   981.3123   13.6934   276.0431
20:097:02100   981.3120   13.6939   276.0435
20:097:02400   981.3117   13.6943   276.0439
20:097:02700   981.3115   13.6947   276.0442
20:097:03000   981.3112   13.6951   276.0446
20:097:03300   981.3109   13.6956   276.0450
20:097:03600   981.3106   13.6960   276.0454
```

Please note that the epoch is written in the format 2-digit year, doy, and second of day. The ZTD solutions are given in the same epoch format. The meteorological parameters must have the exact same epoch, otherwise it will not be read. Furthermore, the file name of the meteorological data must be the exact same as ZTD files (including the extension). For example, if the ZTD file name is `bako1550.20zpd`, the meteorological file should be named `bako1550.20zpd.met`.

2.4 Output Files

All the output files will be saved in the `RES` directory with the format `.pwv`. The file name of the output will be exactly the same as ZTD files (including the extension). For example, if the ZTD file name is `bako1550.20zpd`, the output PWV file will be named `bako1550.20zpd.pwv`.

Below is an example of the `.pwv` file. From left to right: epoch, ZTD in meters, ZWD in meters, and PWV in mm.

Epoch	ZTD(mm)	ZWD(mm)	PWV(mm)
24:015:00000	2563.3000	0.2955	47.6295
24:015:00300	2562.9000	0.2951	47.5650
24:015:00600	2562.7000	0.2949	47.5327
24:015:00900	2562.6000	0.2948	47.5166
24:015:01200	2562.7000	0.2949	47.5327
24:015:01500	2562.7000	0.2949	47.5327
24:015:01800	2562.6000	0.2948	47.5166
24:015:02100	2562.5000	0.2947	47.5004
24:015:02400	2562.4000	0.2946	47.4843
24:015:02700	2562.5000	0.2947	47.5004
24:015:03000	2562.4000	0.2946	47.4843
24:015:03300	2561.9000	0.2941	47.4037
24:015:03600	2562.5000	0.2947	47.5004

3 How to Use ZTD2PWV

Once you have made sure that the program can run properly by following the steps outlined in Section 2.1, ZTD2PWV is ready.

1. Prepare all the input files (ZTD, meteorological data, and `STA_INFO`) and copy them in the appropriate directories.
2. Open your WSL or Linux terminal, then go to your working directory.
3. Run the program by typing `./ZTD2PWV`. Successful run will look like Figure 1.
4. Once it is finished running, check output `.pwv` files in the `RES` directory.

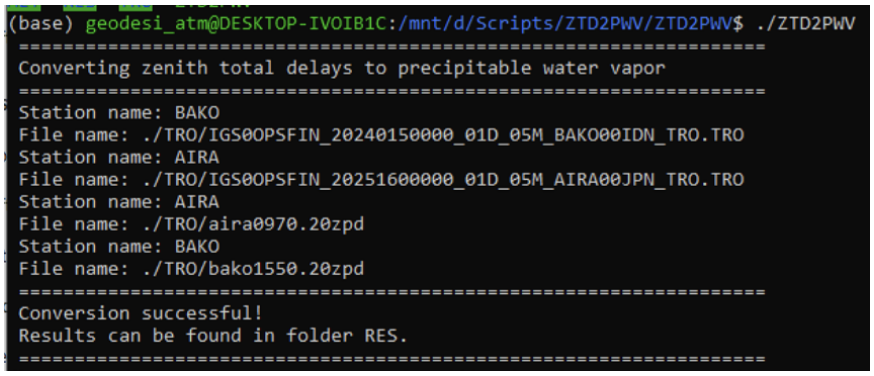

Figure 1: Successful run of ZTD2PWV.

4 Get in Touch

If you encounter any bugs or problems, please send an email to: nabila.sofia@itb.ac.id.

References

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